

A video frame showing a large, green, turbulent wave crashing against a rocky shore. The text "ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK" is overlaid in white, capital letters. In the background, there are faint, semi-transparent images of people's faces and a grid pattern.

ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

Introduction to HPC at ACEnet

Dr. Michelle Shaw, Computational Research Consultant

What is HPC?

From webopedia:

“(n.) A branch of computer science that concentrates on developing supercomputers and software to run on supercomputers. A main area of this discipline is developing parallel processing algorithms and software: programs that can be divided into little pieces so that each piece can be executed simultaneously by separate processors.”

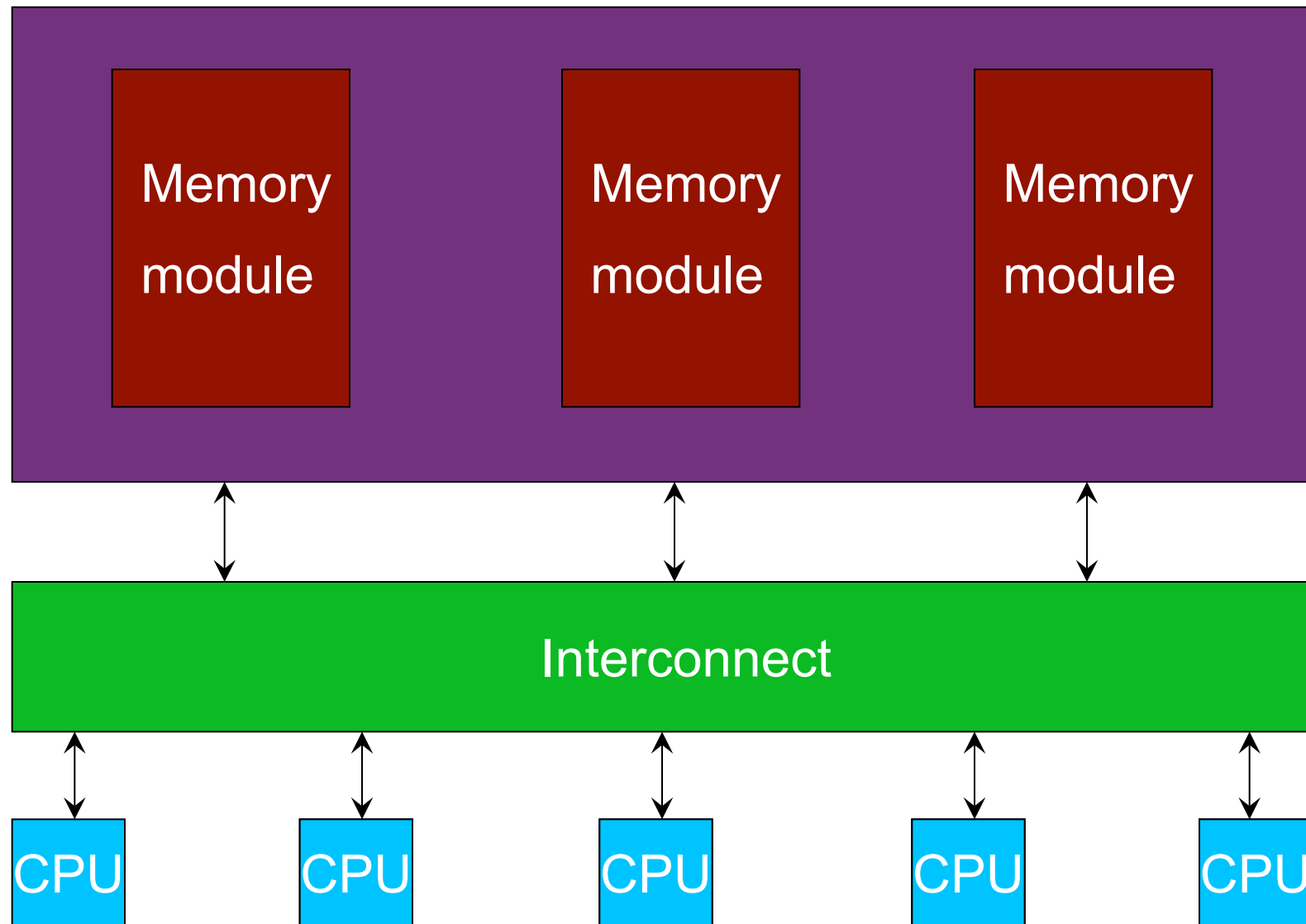
Who Uses HPC?

- Researchers in universities, government and industry
- Research areas utilizing HPC resources are from all disciplines, including:
 - Science (chemistry, physics, biology, mathematics, computer science)
 - Engineering (structural, mechanical, chemical)
 - Social studies and humanities
 - Business
 - Medicine
- Utilized primarily by those that require more compute power than a desktop PC can provide

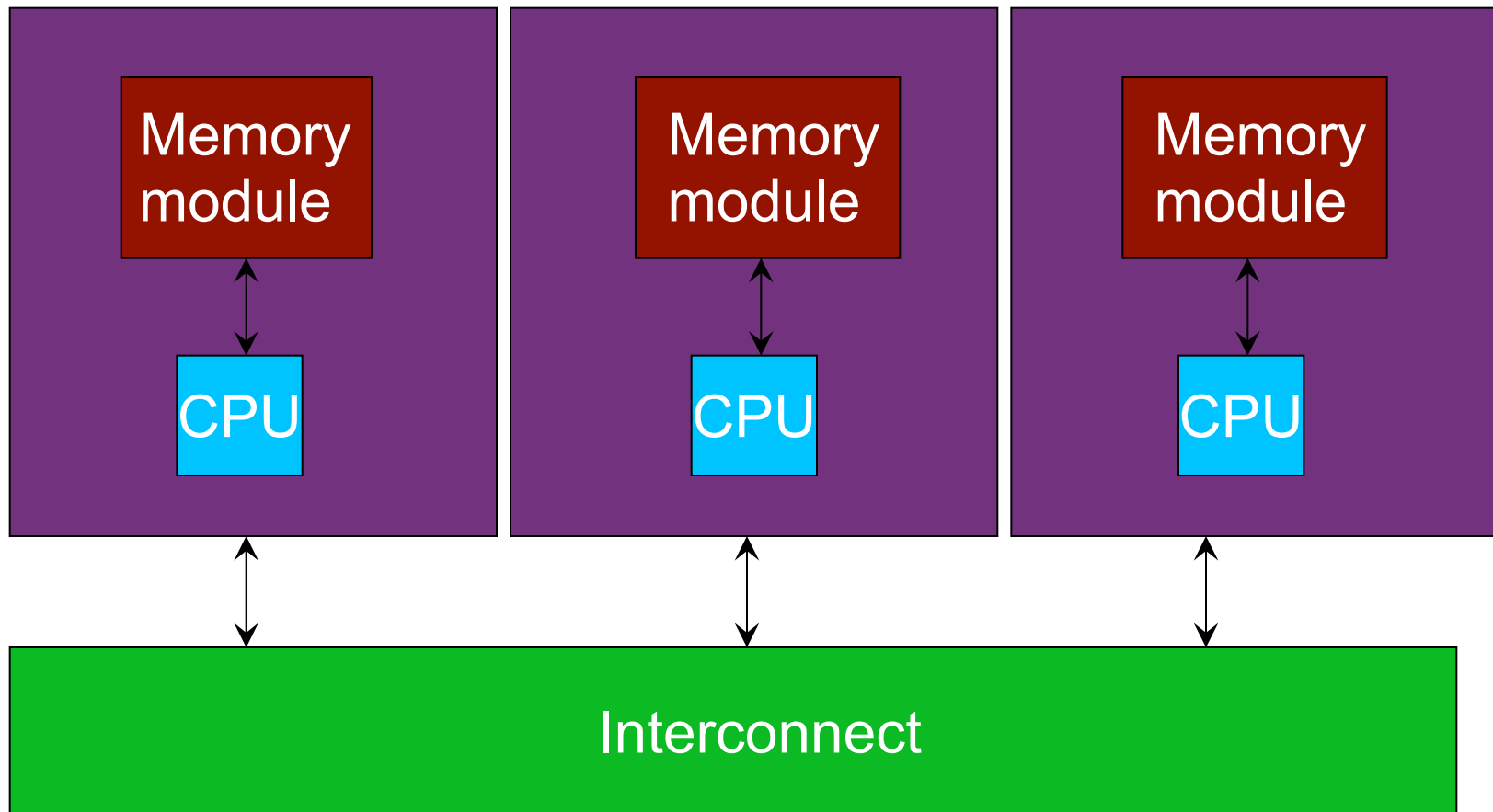
Types of HPC Clusters

- Types of HPC cluster architectures include:
 - Shared memory
 - Distributed memory
 - Hybrid shared/distributed memory
- Uses of these depend heavily on the application and communication/decomposition algorithm

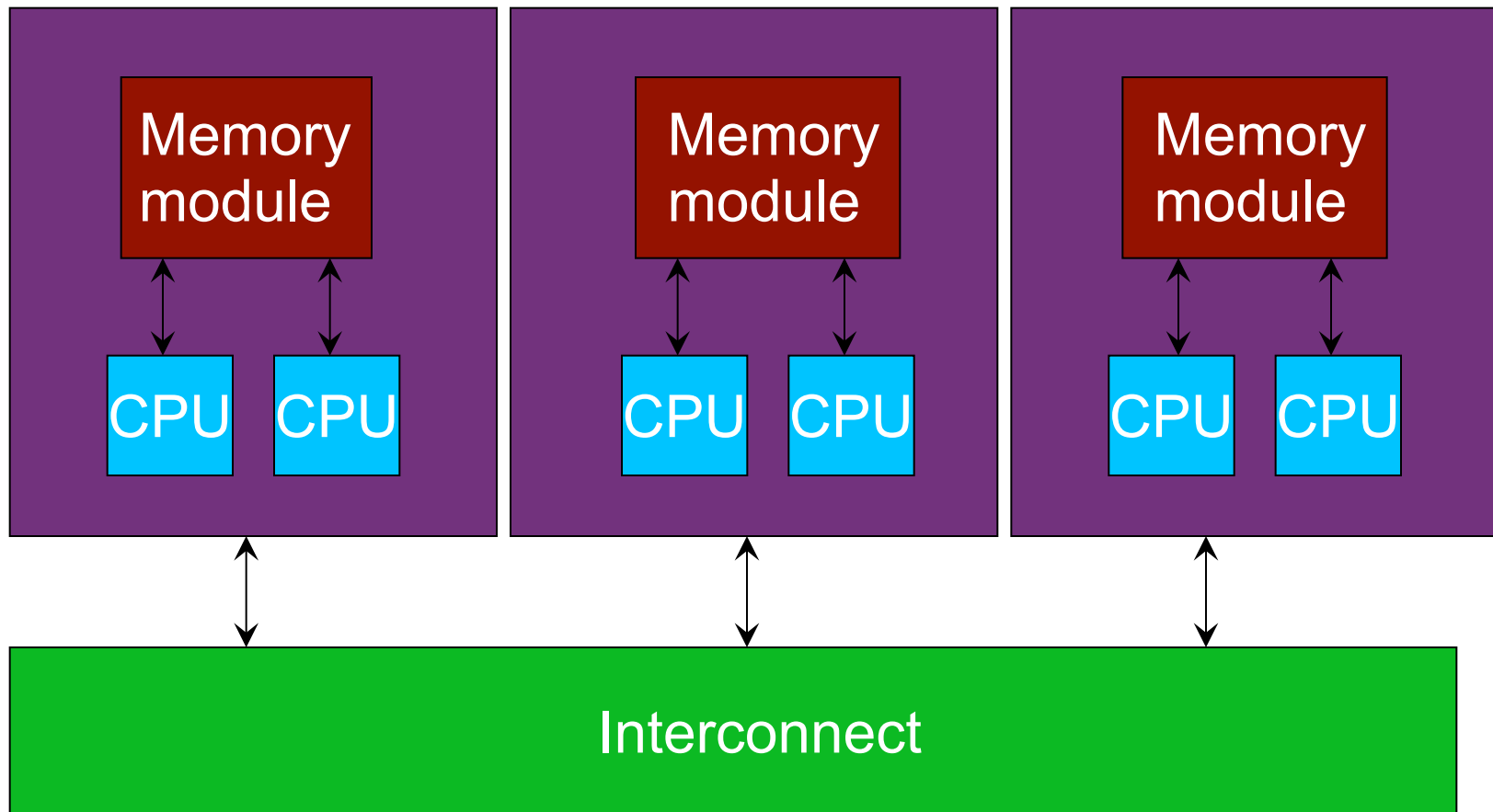
Shared Memory Architecture



Distributed Memory Architecture



Hybrid Architecture



Need for Different Architectures?



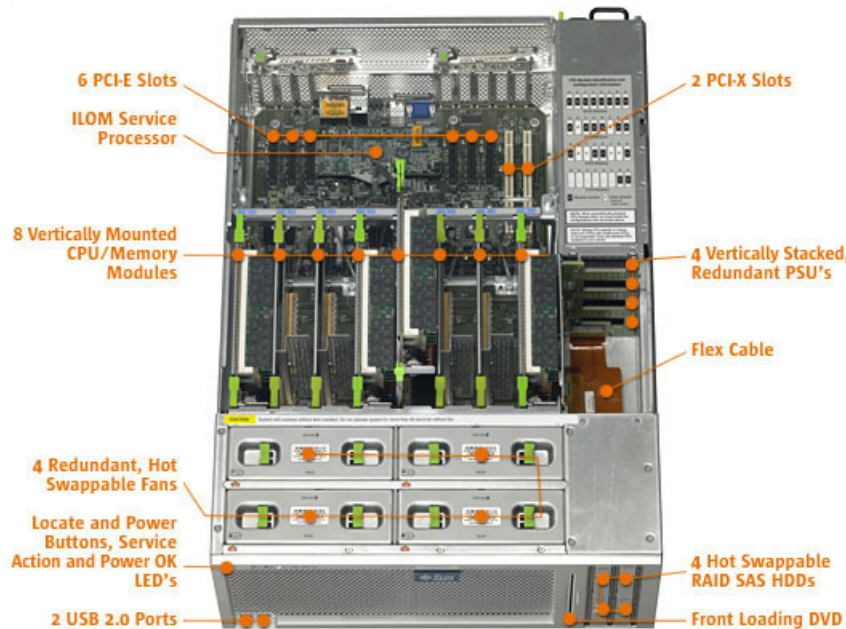
- Some work on HPC clusters is serial, but most is parallel
- Parallel tasks divide up the entire pool of work into pieces to do on each processor
- End result is the same, but the time to complete is reduced significantly
- Different types of parallel algorithms:
 - Distributed memory (or message passing)
 - Shared memory
- Each type of parallel algorithm is designed to exploit the hardware layout

HPC at ACEnet

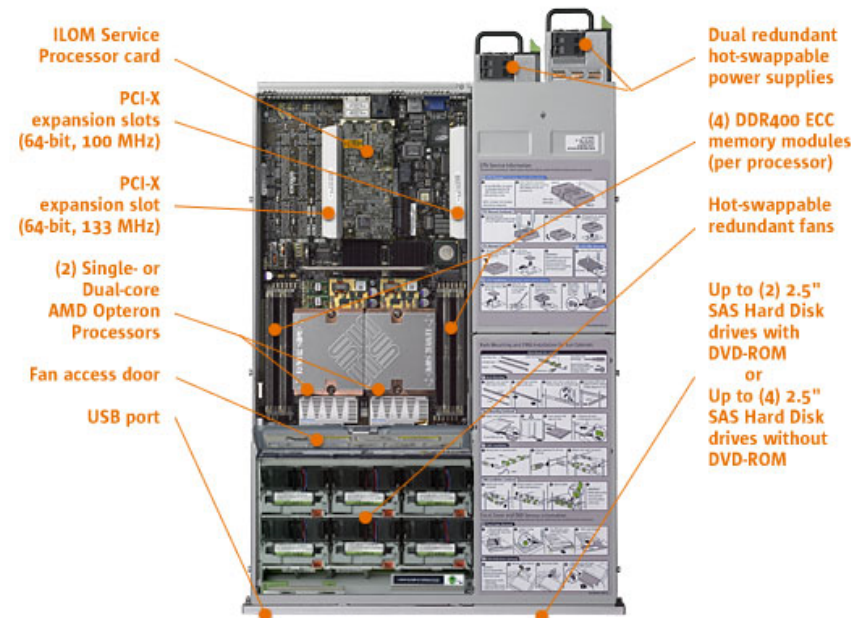


- ACEnet stands for “Atlantic Computational Excellence Network”
- A group of HPC clusters located in Atlantic Canada
- The clusters are of the **hybrid** type, i.e. they contain **nodes** with multiple CPUs connected via an **interconnect**
- Each node has one set of memory shared by all the CPUs on the node

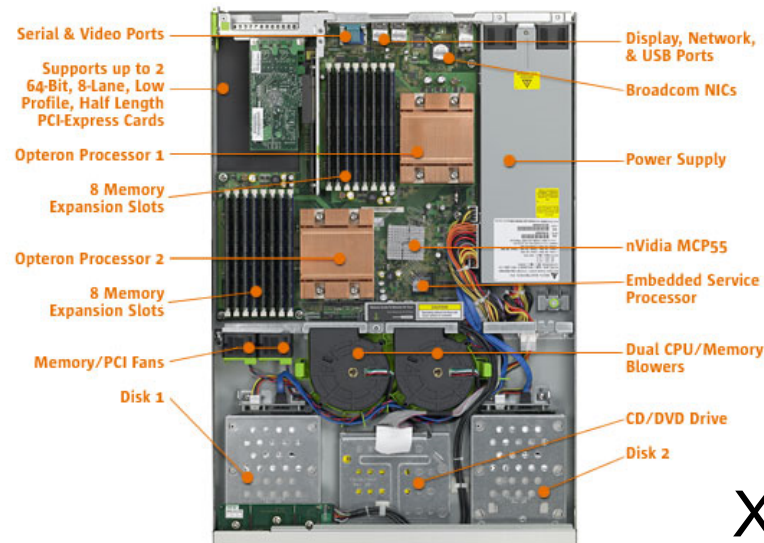
HPC at ACEnet



X4600



X4100



X2200

HPC at ACEnet



- NOTE: though ACEnet has multiple clusters, users have only ONE account
- The login credentials for this account work on all the machines.
- ACEnet has 6 clusters:
 - Brasdor: primarily for serial work
 - Mahone: parallel work needing faster interconnect
 - Placentia: primary cluster for commercial software
 - Fundy: primarily for shared-memory work
 - Glooscap: parallel message passing work
 - Courtenay: primarily for code development and testing

Available Hardware



<i>cluster</i>	<i># cores^{a)}</i>	<i>Operating System^{b)}</i>	<i>RAM/core</i>	<i>Interconnect</i>
brasdor	1152	Linux	2 GB	GigE ^{c)}
fundy	632	Linux	4-8 GB	Infiniband ^{c)}
mahone	556	Linux	4-16 GB	Myrinet ^{d)}
placentia	2680+400	Linux	4 GB	GigE ^{c)} /Infiniband ^{d)}
glooscap	848	Linux	1 GB	GigE ^{c)}
courtenay	52	Linux	4 GB	GigE ^{c)}

a) Processors – 2.6 GHz dual-core Opteron 285 SE

b) Linux – Red Hat Enterprise Linux 4 and 5

c) GigE – Gigabit Ethernet

d) Myrinet – Myrinet 2000 or 10G

[wiki.ace-net.ca/index.php/Compute Overview](http://wiki.ace-net.ca/index.php/Compute_Overview)



Available Storage

<i>Cluster</i>	<i>Total Disk Space - home</i>	<i>Total Disk Space - scratch</i>	<i>Disk Quotas per User – home</i>	<i>Disk Quotas per User – scratch</i>
Brasdor	20 TB	14 TB	47 GB	238 GB
Mahone	20 TB	9 TB	13 GB	238 GB
Placentia	17 TB	9 TB	13 GB	238 GB
Fundy	20 TB	14 TB	47 GB	238 GB
Glooscap	10 TB	7 TB	13 GB	48 GB
Courtenay	2 TB	2 TB	N/A	N/A

[wiki.ace-net.ca/index.php/User Guides](http://wiki.ace-net.ca/index.php/User_Guides)

Code Development Tools



- Compilers and debuggers:
 - Totalview (www.totalviewtech.com)
 - GNU compilers
 - PGI compiler suite (www.pgroup.com)
 - Valgrind
 - Sun compiler suite
- Parallel programming tools:
 - OpenMPI
 - OpenMP (available with compilers)
 - Totalview/PGI debugger (debugs parallel applications)
- Other:
 - Shell scripting: perl, python, ruby, bash, tcsh

Popular Applications



- chemistry:
 - Gaussian (www.gaussian.com)
 - Gamess
 - NWChem
 - DL_POLY
 - NAMD
 - Gromacs
- Physics:
 - Ansys 12 (www.ansys.com)
 - Abinit
 - CPMD
 - Quantum-ESPRESSO
 - ELMER-FEM
 - StarLab
- Math/stats:
 - Matlab (distributed computing engine only)
 - Maple
 - R
 - Maple
- Other packages:
 - MrBayes
 - VMD
 - VTK
 - PLINK

Other Facilities



Internet videoconferencing - inSORS & Access Grid
Dalhousie - Memorial - Mount Allison - Saint Mary's - Saint Francis Xavier
Acadia - New Brunswick + **other sites across Canada & the world**



Other Facilities



Data cave

3D immersive data exploration facility @ Saint Mary's



For Additional Help



- ACEnet User Wiki
wiki.ace-net.ca
- ACEnet User Guide
[wiki.ace-net.ca/index.php/User Guides](http://wiki.ace-net.ca/index.php/User_Guides)
- Research/user support personnel
email: support@ace-net.ca

Connecting to ACEnet Machines



- Logging in
 - ssh *Secure Shell*
- Moving Data Around
 - sftp *Secure File Transfer Protocol*
 - scp *Secure Copy*
- Windows? *Use PuTTY and WinSCP*

Logging in – the Command Line



Generally,

```
> ssh userid@host.ace-net.ca
```

Specifically, today:

```
> ssh userid@mahone.ace-net.ca
```

Logging in - PuTTY



PuTTY Configuration

Category:

- Session
 - Logging
- Terminal
 - Keyboard
 - Bell
 - Features
- Window
 - Appearance
 - Behaviour
 - Translation
 - Selection
 - Colours
- Connection
 - Data
 - Proxy
 - Telnet
 - Rlogin
 - SSH**
 - Serial

Basic options for your PuTTY session

Specify the destination you want to connect to

Host Name (or IP address) Port

Connection type:

☐ Raw ☐ Telnet ☐ Rlogin ☒ **SSH** ☐ Serial

Load, save or delete a stored session

Saved Sessions

Default Settings

Close window on exit:

☐ Always ☐ Never ☒ Only on clean exit



 mahone2.ace-net.ca - PuTTY

login as: dshaw

dshaw@mahone2.ace-net.ca's password:

Last login: Wed Oct 8 11:08:12 2008 from insors2.physics.mun.ca

dshaw@mahone2: ~ > 



Looking Around

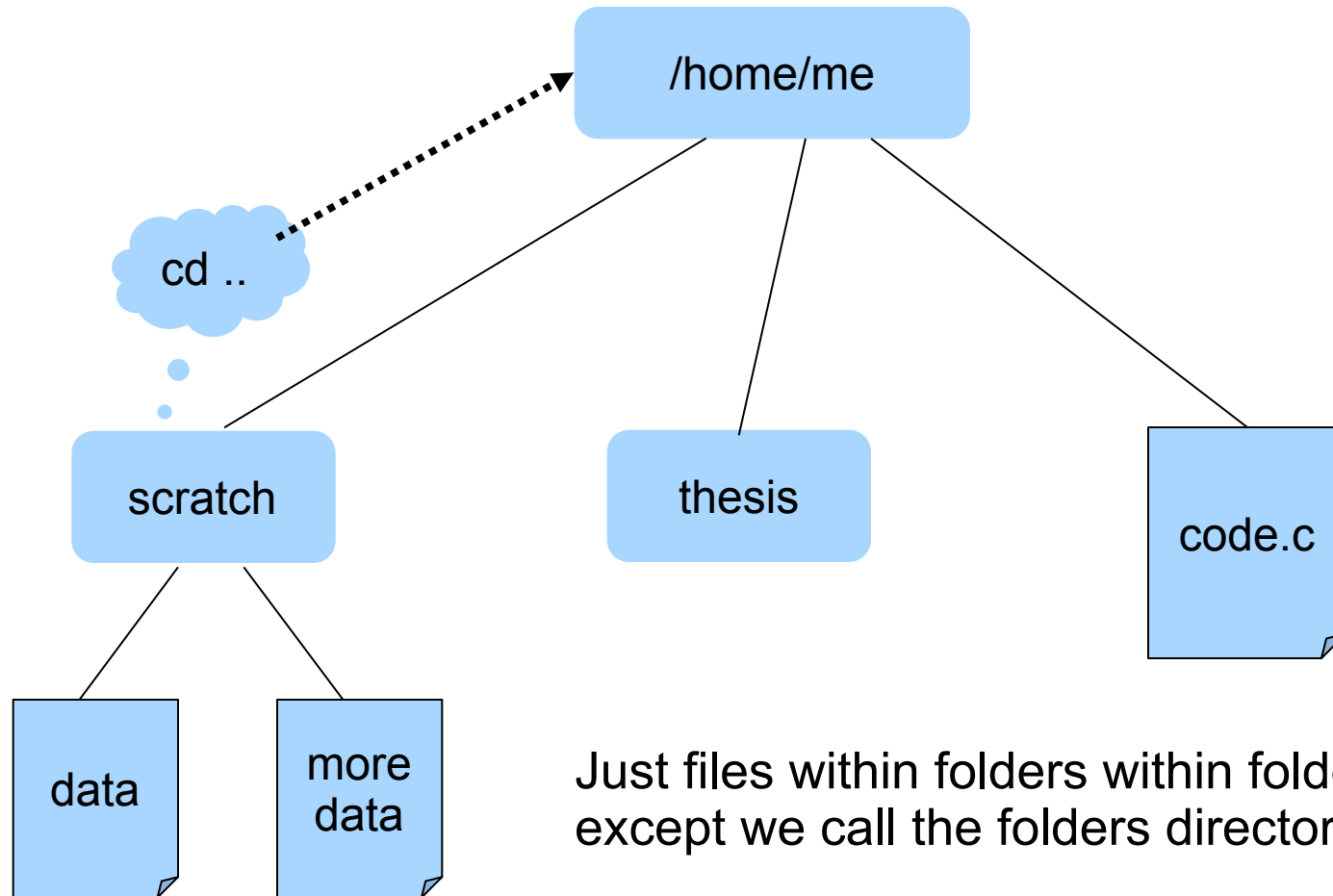
- pwd
 - “print working directory”
 - where am I?
- ls
 - “list”
 - what files are here? what subdirectories?

Getting around

- `mkdir newdir`
 - “make directory”
- `cd directory`
 - “change directory”
- `cd ..`
 - “..” means “parent directory” or “one level up”
 - “.” means “right here”
 - “~” means “my home directory”

Directories grow on trees

Ok, upside down trees...



Just files within folders within folders,
except we call the folders directories.

Moving stuff around

- `mv` *from to*
 - “move”, also “rename”
- `cp` *from to*
 - “copy”
- `rm` *file*
 - “remove”
 - Careful --- There's no recycle bin!
- `rmdir` *directory*

Finding Stuff



- which *program*
 - shows absolute path of *program*
- grep:
 - Search in the file(s) for a string
 - Example: `grep "string" $HOME/file`
- find:
 - Find a file or directory matching a particular parameter
 - Example: `find . -name "foo"`



Paths, directories, abbreviations

- directory separator is “/”
- /home/jsmith/code/test is an *absolute path*
 - begins with “/”
- code/test is a *relative path*
 - no “/” at the beginning
 - same thing if pwd is /home/jsmith
- * (“wildcard”):
 - Put something here

Some Useful Operations

- Some things that are useful on the command line:
 - Pipe: “|”
 - Redirection operators:
 - >, < (redirect stdout and stdin, respectively)
 - >> (similar to redirect, but append if file exists)
 - tab completion – use the tab key to complete names of files/directories/commands
 - & (run command in background)



What's inside?

- more *file*
- less *file*
 - scroll through file, one screen at a time
 - Spacebar to continue, q to quit, ctrl-B to back up
- Editors
 - **vim** *file*
 - **emacs** *file*
 - or use your computer and transfer with sftp

vim

A screenshot of a PuTTY terminal window titled "mahone2.ace-net.ca - PuTTY". The terminal displays C code: "#include <stdio.h>", "main()", "{", " printf ("Hello World!\n");", "}", followed by several tilde (~) characters representing blank lines. At the bottom right, the coordinates "1, 1" and the word "All" are visible next to a vertical scrollbar.

```
#include <stdio.h>

main()
{
    printf ("Hello World!\n");
}
~
~
~
~
~
~
~
~
~
~
~
```

1, 1 All

File Permissions



Sometimes you need to edit the permissions of a file. The permissions of a file dictate who can execute, read and write to a file.

If you do **ls -l** you might see something like the following:

```
-rwxr-xr-x  1 dshaw TechTeam    271 Jul 12  2007 Makefile
```



Second group of 3: What group members can do

Third group of 3: What everyone else can do

First group of 3: what you (owner) can do --- read, write, execute?

- To see a directory listing you must have *execute* permission
- To change permissions, use **chmod**
 - **chmod +x file** to turn on execute permission for you
 - **chmod o-r file** to turn off read permission for others

Managing Your Environment



- which *program*
 - shows absolute path of *program*
- echo \$PATH
 - Print the value of the *environment variable* PATH
- env - shows all set environment variables
- setenv
 - “set environment”
 - shows or sets variables

System Information



- **du** – disk usage
 - How much space is currently being used?
 - Usage: `du -h $HOME`
- **ps** – show the current processes
 - By itself, shows your running processes
 - *ps aux* shows all processes on the system
- **top** – shows running processes
 - Also shows CPU and memory usage for each process
- **date** – gives current date and time

Packing and Unpacking Files

- tar:
 - Pack or unpack a set of files and/or directories into a single file
 - Example: `tar -vcf filename.tar dir/files`
- gzip, gunzip:
 - Compress or uncompress a file
 - Example: `gzip filename.tar`
- Note: the “-z” flag to tar (linux only) can also compress/uncompress files using gzip

Data from the outside world or from one cluster to another



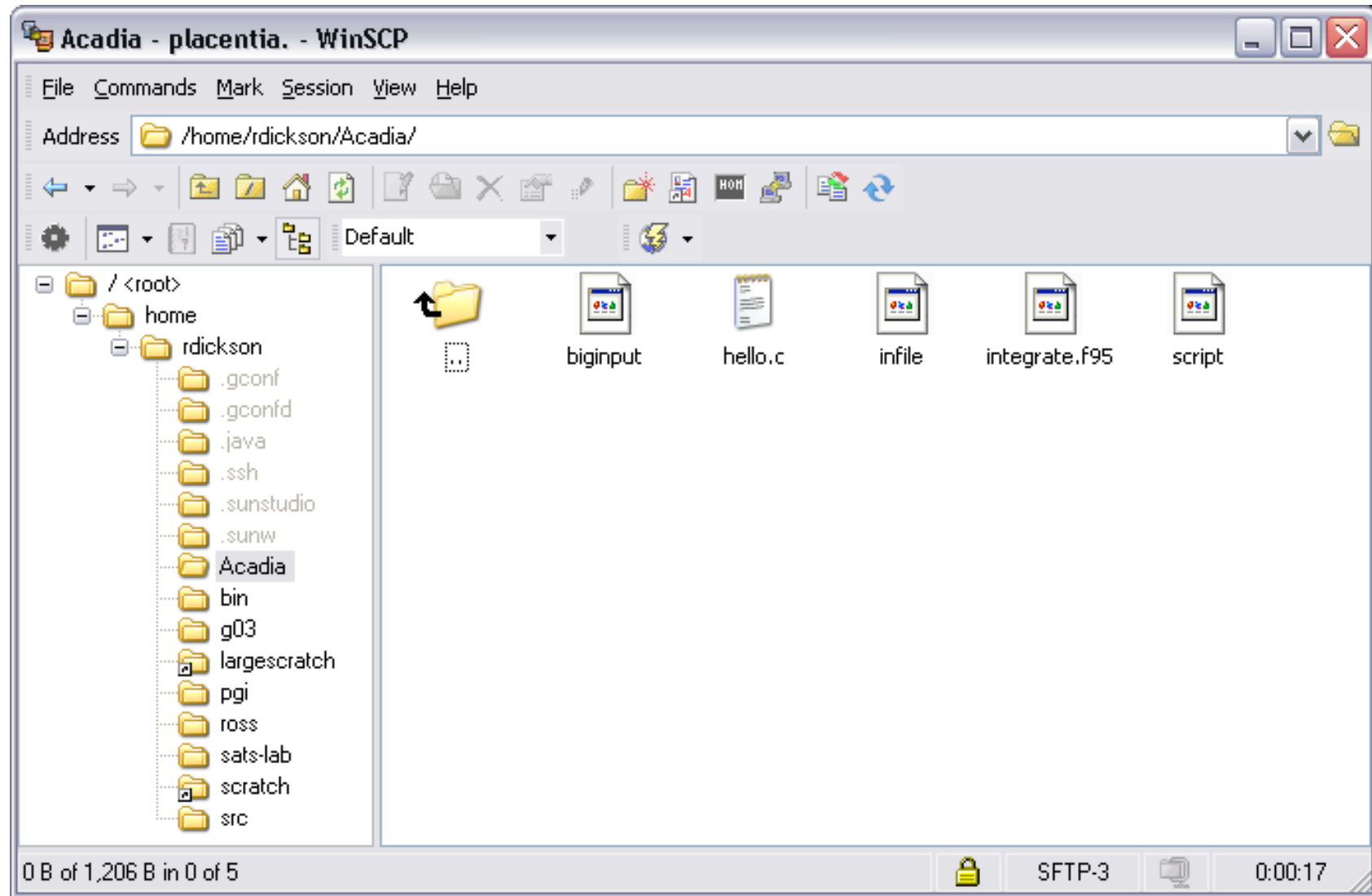
- **sftp** *Secure File Transfer Protocol*
 - provided with Mac OS X and Linux
 - **sftp** username@somehost.ace-net.ca
 - Easiest way to move data between clusters
 - MS Windows? Use WinSCP or Cygwin
 - **scp** Secure Shell Copy
 - scp ...

sftp commands



- ls
 - List files already *there*
- put *filename*
 - Copy file from *here* to *there*
 - Can use wildcards: “put *” copies everything
- get *filename*
- cd *directory*
 - Change directory *there*
- lcd *directory*
 - “Local cd”, change directory *here*

WinSCP





More Linux Info

- `man command`
 - “manual”
- `man -k keyword`
 - searches online manual pages
- `info (coreutils) command`
- Online Linux tutorials
 - Google “Unix command line tutorial”